

LONDON HEATHROW, United Kingdom

WMO: 037720

Lat: 51.4792N

Lon: 0.4506W

Elev: 25

StdP: 101.02

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	-1.1	-6.0	2.3	0.5	-4.6	2.6	1.1	13.1	10.3	11.8	9.8	2.5	270	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	28.7	18.7	26.5	17.8	24.7	17.1	19.8	26.3	18.8	24.3	18.0	22.9	4.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	21.6	16.7	11.9	20.8	16.0	11.4	20.0	56.5	26.3	53.4	24.4	50.7	23.0	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.2	8.2	-4.8	32.2	1.4	2.2	-5.8	33.8	-6.7	35.1	-7.4	36.3	-8.5	37.9		
			-5.3	21.4	1.4	0.8	-6.3	22.0	-7.1	22.5	-7.9	22.9	-8.9	23.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	5.5	5.9	7.8	10.3	13.4	16.6	18.9	18.5	15.8	12.4	8.5	6.0
	DBStd	5.62	3.16	3.13	2.95	3.04	3.06	2.74	2.76	2.58	2.41	2.77	3.07	3.52
	HDD10.0	578	141	117	79	32	5	0	0	0	0	10	64	129
	HDD18.3	2534	397	347	327	241	155	66	26	28	83	185	297	381
	CDD10.0	1187	2	3	11	41	112	198	276	264	173	84	18	6
	CDD18.3	102	0	0	0	0	3	14	44	34	6	1	0	0
	CDH23.3	717	0	0	0	4	28	119	313	215	34	3	0	0
	CDH26.7	173	0	0	0	0	2	27	82	58	4	1	0	0
Wind	WSAvg	4.0	4.4	4.3	4.2	3.8	3.9	3.9	3.8	3.7	3.6	3.9	3.8	4.2
Precipitation	PrecAvg	602	54	36	43	47	49	51	44	51	54	61	58	56
	PrecMax	820	114	92	95	102	102	128	93	150	140	175	151	119
	PrecMin	421	10	2	6	5	4	6	7	0	3	2	8	9
	PrecStd	100	27	21	25	24	26	31	24	31	38	39	35	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	14.8	18.1	23.5	26.0	29.9	31.5	31.2	26.7	21.6	16.4	14.4
		MCWB	11.6	10.9	11.5	14.6	16.8	18.9	19.9	19.9	18.0	16.8	14.4	12.5
	2%	DB	12.4	12.7	15.5	19.8	23.8	26.5	29.0	28.1	23.9	19.1	15.1	13.3
		MCWB	10.8	10.2	10.6	12.9	16.0	17.6	18.7	19.0	16.8	15.3	13.2	11.8
	5%	DB	11.3	11.6	13.8	17.4	21.4	24.2	26.7	25.6	21.7	17.6	14.1	12.5
		MCWB	9.8	9.4	10.0	11.7	14.9	16.6	17.9	17.8	16.0	14.5	12.2	11.0
	10%	DB	10.4	10.5	12.4	15.6	19.1	22.2	24.6	23.5	20.2	16.6	13.1	11.4
		MCWB	8.8	8.6	9.4	10.7	13.5	15.6	17.2	17.0	15.2	13.9	11.3	9.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.7	12.6	15.4	18.4	20.2	21.0	20.9	19.4	17.3	14.9	13.0
		MCDB	13.0	13.4	15.8	21.6	24.1	27.4	29.6	29.0	24.0	20.8	16.1	13.9
	2%	WB	11.0	10.7	11.5	13.6	16.7	18.5	19.7	19.7	18.0	16.1	13.6	12.1
		MCDB	12.2	12.2	14.2	18.6	21.7	24.4	26.5	26.0	22.3	18.1	14.8	13.2
	5%	WB	10.0	9.7	10.7	12.3	15.5	17.3	18.7	18.7	17.0	15.2	12.6	11.2
		MCDB	11.2	11.2	13.1	16.1	20.3	22.4	24.6	23.7	20.1	17.0	13.9	12.3
	10%	WB	9.0	8.9	9.9	11.3	14.3	16.4	17.9	17.8	16.2	14.4	11.5	10.1
		MCDB	10.2	10.3	12.1	14.8	18.4	21.2	23.4	22.0	19.0	16.2	12.9	11.3
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	7.0	8.3	8.6	8.9	9.1	8.7	8.0	6.7	5.6	4.9
		MCDBR	5.7	6.5	9.0	11.1	12.0	12.3	12.5	11.7	10.2	7.5	5.6	4.9
		MCWBR	5.4	4.9	5.4	5.7	5.8	5.5	4.7	4.5	4.7	4.6	4.5	4.6
	5% WB	MCDBR	5.6	5.9	7.9	10.0	10.7	10.7	10.8	10.1	8.5	6.6	5.5	5.0
		MCWBR	5.5	5.0	5.3	5.6	5.6	5.4	4.7	4.6	4.8	4.8	4.9	4.9
Clear-Sky Solar Irradiance	taub	0.332	0.347	0.384	0.406	0.410	0.410	0.405	0.396	0.378	0.361	0.335	0.320	
	taud	2.364	2.369	2.260	2.220	2.250	2.276	2.297	2.341	2.387	2.415	2.409	2.350	
	Ebn at Noon	672	770	802	830	844	846	843	832	805	743	665	618	
	Edh at Noon	63	82	109	127	129	127	123	112	96	77	60	55	
All-Sky Solar Radiation	RadAvg	0.82	1.47	2.55	3.94	4.78	5.29	5.01	4.16	3.14	1.82	0.97	0.64	
	RadStd	0.07	0.19	0.35	0.46	0.30	0.54	0.49	0.34	0.27	0.16	0.11	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (82 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page